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"Catch'em-Bot" Robot Abstract

General Idea:

So, I realize this might not be the most practical thing, but I was watching a kid play catch with his Dad and thought "hey, wouldn't it be cool if I could play catch with a robot?" So, my goal is to make a robot that senses a ping pong ball coming its way and does its best to get into the optimal position to catch the ball.

I'm aware that tracking a moving target like this – especially one that is potentially moving relatively fast – is a difficult software task, but I'm not too shabby at software (if I say so myself), and I think that I could pull it off.

While I'm 99% sure I'll spend the whole semester working on this problem, if by some miracle I finish early, then I thought it would be cool if the robot threw the ball back at you, just like a game of catch. "Better yet," my naive self thought, "what if two robots played catch with each other!" And so was born my design. Again, I would only worry about implementing fun features when the base robot – **a ball catching robot** – was fully functional.

Purpose:

The primary purpose of the robot is detect an incoming ball, roughly calculate where the ball will be in the future, decide where it should be in order to catch said incoming ball, and move to this destination. I imagine that this could be done for recreation, giving hardworking engineers the boost to their day they need to do more "important" things.

Should the ball catching mechanism be fully and successfully implemented, the robot will also, upon receiving the ball, detect a symbol, determine at what angle it should throw the ball to successfully reach said signal, reposition itself if necessary, and shoot the ball towards that symbol.

Function:

The robot will function autonomously once activated by a given user. Once it has successfully caught its target ball, it will wait (or potentially roam around) until it restarts (either by detecting the ball has been removed or some other external stimulus, such as a user requested reset).

Structure/Design:

A hideous and child-like sketch of the robot's early, early design is given below:

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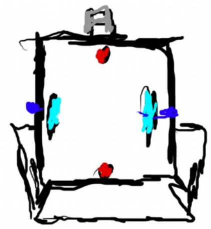


I cannot draw to save my life
Things are horribly not to scale

DC motors (brushless?) powering wheels/treads. Their location is very temporary; I don't know Mechanical stuffs yet. Controlled by a uC in

Casters or somesuch wheel-like thing. Not powered by motors.

Motors (brushed DC?) powering the ball shooter. Controlled by

~~Sign. Used by paired bot to identify its partner bot.~~
(This is, of course, if I can somehow manage to make 2 robots. Of course, I will make 1 fully functional before I even think about implementing 2)

Servo used to angle the shooter/camera. Controlled by . Obviously don't look like joints or circles; I was just highlighting what item would move

Electronics. Potentially 4-5 ESCs or motor controllers or what-have-you for all motors. uCs (TI launchpad) to control motors; brains are Jetson TK1, which also does OpenCV. Also Voltage Regulators, probably other stuff, too? I'm new at this...

IR sensors that talk to , which may influence in order to avoid objects/ detect obstacles between paired bots. (May replace with ultrasound sensors or something else?)

Camera. Feeds information to Jetson TK1 in . Will have to be fast (i.e., maybe not USB); used to both find target and catch incoming target ball. (Potentially does other stuff?)

Ball loading mechanism. Takes balls from the chassis bin and loads them into the firing mechanism (I'm imagining like a conveyer belt type thing. There's probably a simpler way, too. Will have to wait until given a signal from to finally push the balls into where they are pushed by

Chassis. Includes basket that holds its target balls as they fall to the loading mechanism.
(I'll be the first to admit that this sort of design is not my strong suit :)

LCD Screen/Array of LEDs/Keypads/etc. Displays information to user about the current state of the bot. Given this information from

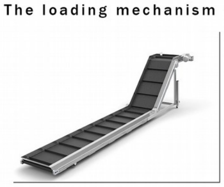
Battery/misc things. This space reserved for the battery/any additional parts that I am currently forgetting.

Some More about the "shooting" mechanism:

I'm thinking something like a baseball pitching machine:

Except horizontal and with ping-pong balls. If that doesn't work as well, I'd do vertical, I guess.

The loading mechanism would be like a conveyor belt I guess?



And there'd be a little arm or something that blocked the ball before it reached the shooting wheels when it gets to the top

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Very very tentative and vague Bill of Materials:

Quantity	Item	Purpose
1-2	Brushless DC Motors	Power the wheels
2	Casters	movement
1-2	Additional Motors (Brushed DC?)	(powering the shooting mechanism)
2	Servos	(angling the camera/shooting thing)
1	Jetson TK1 (I have one already)	(the brains; OpenCV)
4-5	ESCs/Motor Controllers	(control the motors)
1-2	TI Launchpad	(uC, interfaces sensors/motors)
A lot probably	Voltage Regulators	(get power to all devices)
4-8	IR sensors/Bump Switches/etc	(obstacle avoidance)
1	Camera	(ball detection; needs to be fast, hopefully not USB)
1	Conveyor Belt	(Load balls. I realize “conveyor belt” is a bit generic)
		(it could be like, a bunch of gears, like an escalator, etc)
1	LCD Screen or something	(For user feedback)
1	Battery	(Power. Probably LiPo)
1	Basket	(Holding the balls)
4	Wheels	(2 for movement; 2 different ones for the shooting)
Some	Poles or something	(Hold up shooting Mechanism above electronics/battery)
Maybe some	Felt or other soft material	(Catch the balls and make sure they don't bounce out)
Some	Duct Tape	(Hold the universe together)
1	Tube or something	(Aim the ball? Alternatively I guess you could angle the
		shooting wheels and everything, but…)

Wishful Thinking Schedule:

Week of	What I'll get Done Hopefully
01/11/16	Order sensors/camera/motors, learn OpenCV
01/19/16	Affix Motors/uC/some sensors to base. Have “movable” thing
01/25/16	Begin Coding actual logic into bot. Maybe do camera stuff?
02/01/16	Finalize chassis design. Continue debugging
02/08/16	Hopefully OpenCV code working. Get robot to catch things
02/15/16	Debug anything. Begin implementing Shooter Mechanism if time
02/22/16	Debugging
03/07/16	Debugging. If time, look into building partner robot
03/14/16	Debugging
03/21/16	Debugging. Decide on final features, cut anything that hasn't worked
03/28/16	Debugging. If time, make the robot look all pretty
04/04/16	Debugging

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Hopeful Final Robot Capabilities:

In a perfect world, two “Catch'em-bots” could play a game of catch with each other: one robot would start with the ball and search for the symbol held by the other robot. This other robot would be on the lookout for an incoming ball. When the first robot determines it is ready, it will shoot the ball, which will be eagerly caught by the second robot. The robots now switch places, and the process repeats.

Should an obstacle get in the way of the two robots (say, a monkey in the middle), then the shooting robot would ideally find a way to get around to the other robot – either by moving around the obstacle or shooting above the obstacle. Additionally, both robots should exhibit basic object avoidance; in this way, the “game” could be made interesting by causing the receiving robot to move away and the throwing robot to chase after it as it attempts to make a clear shot.

From a broader perspective, the robot(s) will ideally be fun to watch/experiment with, and will “evoke human emotions” as my brother put it. I would consider the project a resounding success if seeing the robot(s) operate brought a smile to someone's face.